Prompt: Everything you do in a day.
Generated SRL Sequence:
<V> woke up <A1> ent0 <A2> to the sound of my alarm blaring
<V> rolled over <A1> ent0
<V> looked <A0> ent0 <A1> at my alarm clock
<V> went <A0> ent0 <A4> to the bathroom
<V> wash <A0> ent0 <A2> myself <A1> of my clothes
<V> got <A0> ent0 <A1> dressed <A2> in my boxers
<V> dressed <A0> by my boxers <A1> ent0 <A2> as ent0 walked to the kitchen
<V> walked <A0> ent0
<V> turned on <A0> ent0 <A1> the water
<V> poured <A0> ent0 <A1> a cup of coffee <A3> in
<V> sat down <A1> ent0

Figure 7: Example *generated* action plan for the SRL + NER Entity Anonymization model. It shows a plausible sequence of actions for a character.

Model	# Unique Verbs	% Diverse Verbs
Human Stories	34.0	76.5
Fusion	10.3	61.1
Summary	12.4	60.6
Keyword	9.1	58.2
Compression	10.3	54.3
SRL	14.4	62.5
+ verb-attention	15.9	64.9

Table 2: Action Generation. Generating the SRL structure improves verb diversity and reduces repetition.

els must give the true mention the highest likelihood. We analyze accuracy on the first mention of an entity, an assessment of novelty, and subsequent references, an assessment of consistency.

Effect of Sub-word Modelling Table 3 shows that modelling a character-level vocabulary for entity generation outperforms BPE and word-based models, because of the diversity of entity names. This result highlights a key advantage of multi-stage modelling: the usage of specialized modelling techniques for each sub-task.

Effect of Additional Context Entity references should be contextual. Firstly, names must be appropriate for the story setting—*Bilbo Baggins* might be more appropriate for a fantasy novel. Subsequent references to the character may be briefer, depending on context—for example, he is more likely to be referred to as *he* or *Bilbo* than his full name in the next sentence.

We compare three models ability to fill entities based on context (using coreference-anonymization): a model that does not receive the story, a model that uses only leftward context (as in Clark et al. (2018)), and a model with access to the full story. We show in Table 3 that having access to the full story provides the best performance. Having no access to any of the story decreases ranking accuracy, even though the model still receives the local context window of the entity as input. The left story context model performs better, but looking at the complete story provides additional gains. We note that full-story context can only be provided in a multi-stage generation approach.

Qualitative Examples Figure 8 shows examples of entity naming in three stories of different genres. We evaluate different genres to examine if generated entities adapt to the style of the story. We show that models can adapt to the context—for example generating *The princess* and *The Queen* when the context includes *monarchy*.

6.4 Effect of Entity Anonymization

To understand the effectiveness of the entity generation models, we examine their performance by analyzing generation diversity.

Diversity of Entity Names Human-written stories often contain many diverse, novel names for people and places. However, these tokens are rare and subsequently difficult for standard neural models to generate. Table 4 shows that the fusion model and baseline decomposition strategies generate very few unique entities in each story. Generated entities are often generic names such as *John*.

Our proposed decompositions generate substantially more unique entities than strong baselines. Interestingly, we found that using coreference resolution for entity anonymization led to fewer unique entity names than generating the names independently. This result can be explained by the coreference-based model re-using previous names more frequently, as well as using more pronouns.

Coherence of Entity Clusters Well structured stories will refer back to previously mentioned characters and events in a consistent manner. To evaluate if the generated stories have these characteristics, we examine the coreference properties in Table 5. We quantify the average number of coreference clusters and the diversity of entities within each cluster (e.g. the cluster *Bilbo, he, the hobbit* is more diverse than the cluster *he, he, he*).

Our full model produces more non-singleton coreference chains, suggesting greater coherence, and also gives different mentions of the same entity more diverse names. However, both numbers are still lower than for human generated stories, indicating potential for future work.

Model	First Mentions			Subsequent Mentions		
	Rank 10	Rank 50	Rank 100	Rank 10	Rank 50	Rank 100
Word-Based	42.3	25.4	17.2	48.1	38.4	28.8
BPE	48.1	20.3	25.5	52.5	50.7	48.8
Character-level	64.2	51.0	35.6	66.1	55.0	51.2
No story	50.3	40.0	26.7	54.7	51.3	30.4
Left story context	59.1	49.6	33.3	62.9	53.2	49.4
Full story	64.2	51.0	35.6	66.1	55.0	51.2

Table 3: Accuracy at choosing the correct reference string for a mention, discriminating against 10, 50 and 100 random distractors. We break out results for the first mention of an entity (requiring novelty to produce an appropriate name in the context) and subsequent references (typically pronouns, nominal references, or shorter forms of names). We compare the effect of sub-word modelling and providing longer contexts.

Model	# Unique Entities
Human Stories	2.99
Fusion	0.47
Summary	0.67
Keyword	0.81
Compression	0.21
SRL + NER Entity Anonymization	2.16
SRL + Coreference Anonymization	1.59

Table 4: Diversity of entity names. Baseline models generate few unique entities per story. Our decompositions generate more, but still fewer than human stories. Using coreference resolution to build entity clusters reduces diversity here—partly due to re-using existing names more, and partly due to greater use of pronouns.

Model	# Coref Chains	Unique Names per Chain
Human Stories	4.77	3.41
Fusion	2.89	2.42
Summary	3.37	2.08
Keyword	2.34	1.65
Compression	2.84	2.09
SRL + NER Entity Anonymization	4.09	2.49
SRL + Coreference Anonymization	4.27	3.15

Table 5: Analysis of non-singleton coreference clusters. Baseline models generate very few different coreference chains, and repetitive mentions within clusters. Our models generate larger and more diverse clusters.

Qualitative Example Figure 9 displays a sentence constructed to require the generation of an entity as the final word. The fusion model does not perform any implicit coreference to associate the *allergy* with *his dog*. In contrast, coreference entity fill produces a high quality completion.

7 Related Work

Decomposing natural language generation into several steps has been extensively explored (Reiter and Dale, 2000; Gatt and Krahmer, 2018). In classical approaches to text generation, various stages were used to produce final written text. For example, algorithms were developed to determine content and discourse at an abstract level, then sentence aggregation and lexicalization, and finally steps to resolve referring expressions (Hovy, 1990; Dalianis and Hovy, 1993; Wahlster et al., 1993; Ratnaparkhi, 2000; Malouf, 2000). Our work builds upon these approaches.

Story Generation with Planning Story generation using a plan has been explored using many different techniques. Traditional approaches organized sequences of character actions with hand crafted models (Riedl and Young, 2010; Porteous and Cavazza, 2009). Recent work extended this to modelling story events (Martin et al., 2017; Mostafazadeh et al., 2016), plot graphs (Li et al., 2013), plot summaries (Appling and Riedl, 2009), story fragments or vignettes (Riedl, 2010), or used sequences of images (Huang et al., 2016) or descriptions (Jain et al., 2017).

We build on previous work that decomposes generation. Xu et al. (2018) learn a skeleton extraction model and a generative model conditioned upon the skeleton, using reinforcement learning to train jointly. Zhou et al. (2018) train a storyline extraction model for news articles, but require supervision from manually annotated storylines. Yao et al. (2019) use RAKE (Rose et al., 2010) to extract storylines, and condition upon the storyline to write the story using dynamic and static schemas that govern if the storyline can change.

Entity Language Models An outstanding challenge in text generation is modelling and tracking entities. Centering (Grosz et al., 1995) gives a theoretical account of how referring expressions for entities are chosen in discourse context. Named entity recognition has been incorporated into language models since at least Gotoh et al. (1999), and can improve domain adaptation (Liu and Liu,

Science Fiction
The captain finger went directly to the Y in the touchscreen keyboard. He rolled his eyes when it activated. It was the part he hated the most of his job. As the man dragged the probe to the room, it started closing the doors and the large room. After he was done, he entered the dock to inspect the bounty he had snatched from the skies. It was a spaceship [...] The first thing he did was stretch his neck, as if in awe of the size of the probe. Such a big device for such a simple message. Obviously the probe was old. And then he saw a small, green thing.

Historical Fiction
Long ago, the tribe sprawled across the barbarian lands of the north. Of the largest, the King of North Island who held lands spanning around the world. The world that was, clashing with people that marshalled their people with song and steel at their back. In years past they were armed with ferocious weapons [...] Now we hold only fragmented relics; relics of such power that the king will not suffer their legions to use, for the untold destruction they brought to the Ancient peoples.

Realistic Fiction
Another Winter festival accounted for. The human race looked marvelous and her speeches became more and more powerful each year. The princess fed on the millennials hopes and dreams. Making her stronger for the new year. Ever since the world event, she has been losing her life force. The millennials are fed up with the baby boomers and have decided a revolt. The Queen doesn't know when or where but her days have now been numbered. People are realizing the monarchy is obsolete and they are renouncing their devotion from her.

Figure 8: Generating entity references for different genres, using entity-anonymized human written stories. Models use the story context to fill in relevant entities. Color indicates coreferent clusters.

John brought his dog to her house, but when she pet him and sneezed, he realized he forgot that she was allergic to ____.

Fusion model:	tuna
Coref Entity Fill:	the dog

Figure 9: Constructed sentence where the last word refers to an entity. The coreference model is able to track the entities, whereas the fusion model relies heavily on local context to generate the next words.

2007). Language models have been extended to model entities based on information such as entity type (Parvez et al., 2018). Recent work has incorporated learning representations of entities and other unknown words (Kobayashi et al., 2017), as well as explicitly model entities by dynamically updating these representations to track changes over time and context (Ji et al., 2017). Dynamic updates to entity representations are used in other story generation models (Clark et al., 2018).

Non-Autoregressive Generation Our method proposes decomposing left-to-right generation into multiple steps. Recent work has explored non-autoregressive generation for more efficient language modeling and machine translation. Ford et al. (2018) developed two-pass language models, generating templates then filling in words. The partially filled templates could be seen as an intermediary representation similar to generating a compressed story. Other models allow arbitrary order generation using insertion operations (Gu et al., 2019; Stern et al., 2019) and Gu et al. (2017) explored parallel decoding for machine translation. In contrast, we focus on decomposing generation to focus on planning, rather than efficient decoding at inference time.

8 Conclusion

We proposed an effective method for writing short stories by separating the generation of actions and entities. We show through human evaluation and automated metrics that our novel decomposition improves story quality.

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